

# Tut-11 DSP ECE/EEE F434

## Finite Word length effects in Digital filters ①

1. Concept of fixed, floating point representation, overflow, and truncation, rounding errors.

2. IEEE 754 standard (Mantissa and exponent concept)

For 32 bit machine

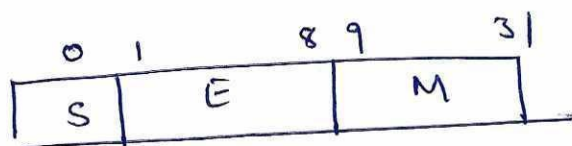
$$X = (-1)^S \cdot 2^{E-127} (M) \quad \text{if } 0 < E < 255$$

$$X = (-1)^S 2^{E-127} (1.M) \quad \text{then}$$

if  $E=0, M \neq 0$ , then

$$X = (-1)^S 2^{-126} (0.M)$$

Where  $0.M \rightarrow$  fraction and  $1.M \rightarrow$  Mixed number with one integer bit & 23 fractional bits



3. Quantization error  
 $x(n) \rightarrow$  original signal (DT),  $x_q[n] \rightarrow$  quantized digital signal  
 $Q[\cdot] \rightarrow$  Quantization operator

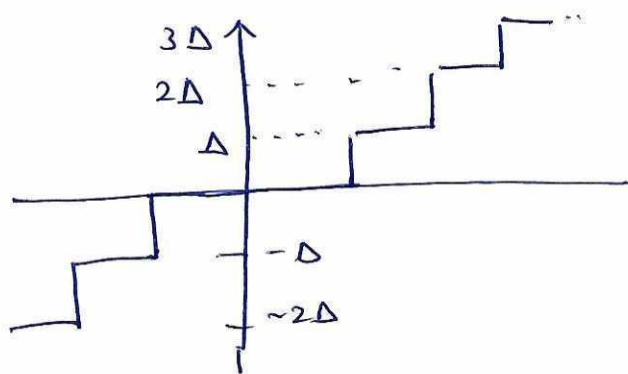
$$x_q[n] = Q[x(n)]$$

$$e_q[n] = x_q[n] - x[n]$$

Uniform pdf of error

as errors (rounding) will be uniformly distributed due to uniform quantization

Uniform quantization



$$\text{Mean} = \int_{-\infty}^{\infty} e \cdot p(e) de = \int_{-\Delta/2}^{\Delta/2} e \cdot \frac{1}{\Delta} de = 0$$

where  $p(e) = \begin{cases} \frac{1}{\Delta}, & |e| \leq \Delta/2 \\ 0, & \text{otherwise} \end{cases}$

$$\text{Variance} = \int_{-\infty}^{\infty} (e - \underbrace{\mathbb{E}[e_q]}_0)^2 p(e) de = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} e^2 de = \frac{\Delta^2}{12} \quad (2)$$

$$\text{Binary} \Rightarrow \Delta = 2^{-b} \Rightarrow \text{var} = \boxed{\frac{2^{-2b}}{12}}$$

Autocorrelation

$$\begin{aligned} \underline{m \neq 0} \quad R_{eq}(m) &= \mathbb{E} \left\{ (e_q(n) - \underbrace{\mathbb{E}[e_q(n)]}_{=0}) (e_q(n+m) - \underbrace{\mathbb{E}[e_q(n+m)]}_{=0}) \right\} \\ &= \mathbb{E}[e_q(n)] \mathbb{E}[e_q(n+m)] = 0 \end{aligned}$$

$$\text{for } \underline{m=0} \quad R_{eq}(m) = \text{Variance}$$

$$\therefore R_{eq}(m) = \begin{cases} \frac{\Delta^2}{12} = \frac{2^{-2b}}{12} ; & m=0 \\ 0, & m \neq 0 \end{cases}$$

$$\Rightarrow \boxed{\frac{\Delta^2}{12} \delta(m)}$$

Consider  
LTI filter

$$x[n] \rightarrow \boxed{h[n]} \rightarrow y[n]$$

effect of uniform quantization noise of I/P signal  
o/p Quantization

$$\begin{aligned} x_q[n] &\rightarrow \boxed{h[n]} \rightarrow y_q[n] \\ &= x[n] + e_q[n] \quad \quad \quad = y[n] + f_q[n] \end{aligned}$$

from linearity property

# O/P quantization noise statistics

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$$e_q[n] \rightarrow [h[n]] \rightarrow f_q[n] = e_q[n] * h[n]$$

$$\begin{aligned} \text{Mean } \{f_q[n]\} &= E[e_q[n] * h[n]] = E\left[\sum_{k=-\infty}^{\infty} e_q[k] h[n-k]\right] \\ &= \sum_{k=-\infty}^{\infty} E[e_q[k]] h[n-k] = 0 \end{aligned}$$

$$\text{Variance } E[(f_q[n]) - (\underbrace{E[f_q[n]]}_{=0})^2] = E[(f_q[n])^2]$$

$$= E\left[\left(\sum_{k=-\infty}^{\infty} e_q[k] h[n-k]\right)^2\right]$$

$$= E\left[\sum_{k=-\infty}^{\infty} e_q[k] h[n-k] \sum_{l=-\infty}^{\infty} e_q[l] h[n-l]\right]$$

$$= E\left[\sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \underbrace{e_q[k] e_q[l]}_{E[e_q[k] e_q[l]] = R_{eq}(l-k)} h[n-k] h[n-l]\right]$$

$$= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \frac{\Delta^2}{12} \delta(l-k) h[n-k] h[n-l]$$

$$= \sum_{k=-\infty}^{\infty} \sum_{l \neq k} \frac{\Delta^2}{12} \delta(l-k) h[n-k] h[n-l] + \sum_{k=-\infty}^{\infty} \sum_{l=k} \frac{\Delta^2}{12} \delta(l-k) h[n-k] h[n-l]$$

$$= \sum_{k=-\infty}^{\infty} \frac{\Delta^2}{12} \cdot 1 \cdot h[n-k] h[n-k]$$

$$= \frac{\Delta^2}{12} \sum_{k=-\infty}^{\infty} h^2[n-k]$$

or in general

$$\text{Var} = \left[ \frac{\Delta^2}{12} \sum_{n=-\infty}^{\infty} h^2(n) \right]$$

Also,  $\sum_{n=-\infty}^{\infty} h^2(n) \leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(\omega)|^2 d\omega$  (4)

Parsavals relation.

For b-bit unsigned fixed pt fractional representations

$$\Delta = 2^{-b}$$

$$\text{Mean} = 0$$

$$\text{Var} = \frac{\Delta^2}{12} \sum_{n=-\infty}^{\infty} h^2(n) = \frac{2^{-2b}}{12} \sum_{n=-\infty}^{\infty} h^2(n)$$

4.  $\text{SNR} = \frac{\sigma_y^2}{\sigma_q^2} \rightarrow \text{signal power}$   
 $\sigma_q^2 \rightarrow \text{quantization error power}$

'SNR' (also used)

Var. of o/p noise  $\sigma_q^2 = \sigma_e^2 \sum_{k=-\infty}^{\infty} h^2(k)$  } Parallel Connection

5. Cascade connection

Find the variance of round off noise at o/p of two cascade realization of filter with system function

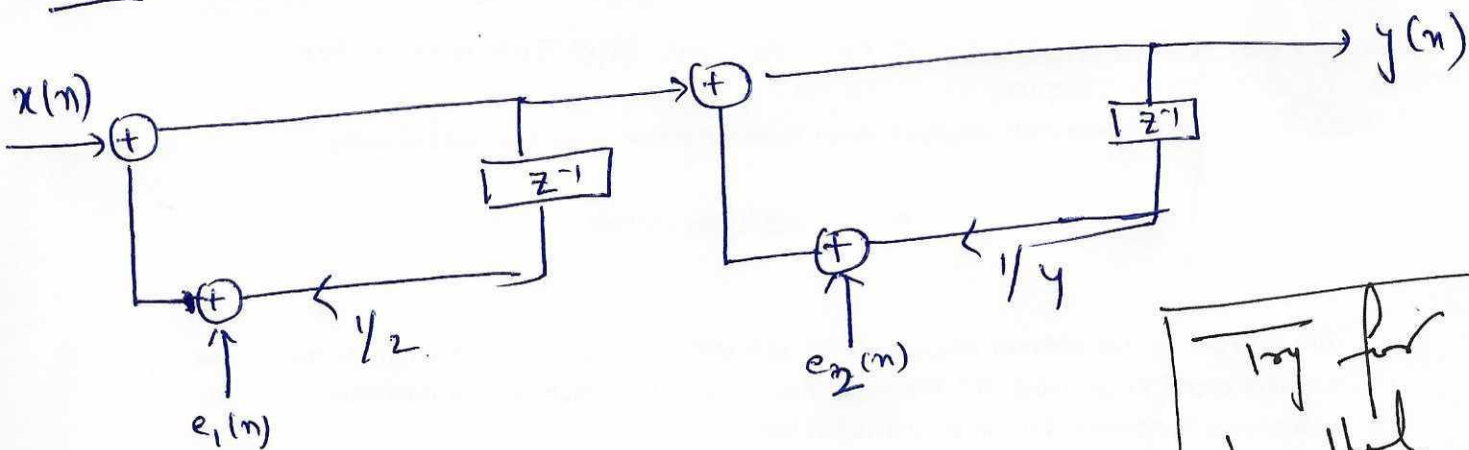
$H(z) = H_1(z) \cdot H_2(z)$  where  $H_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$  and

$H_2(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}$

Sol  $h_1(n) = \left(\frac{1}{2}\right)^n u(n)$   
 $h_2(n) = \left(\frac{1}{4}\right)^n u(n)$   
 $\Rightarrow h(n) = \left[ 2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n \right] u(n)$

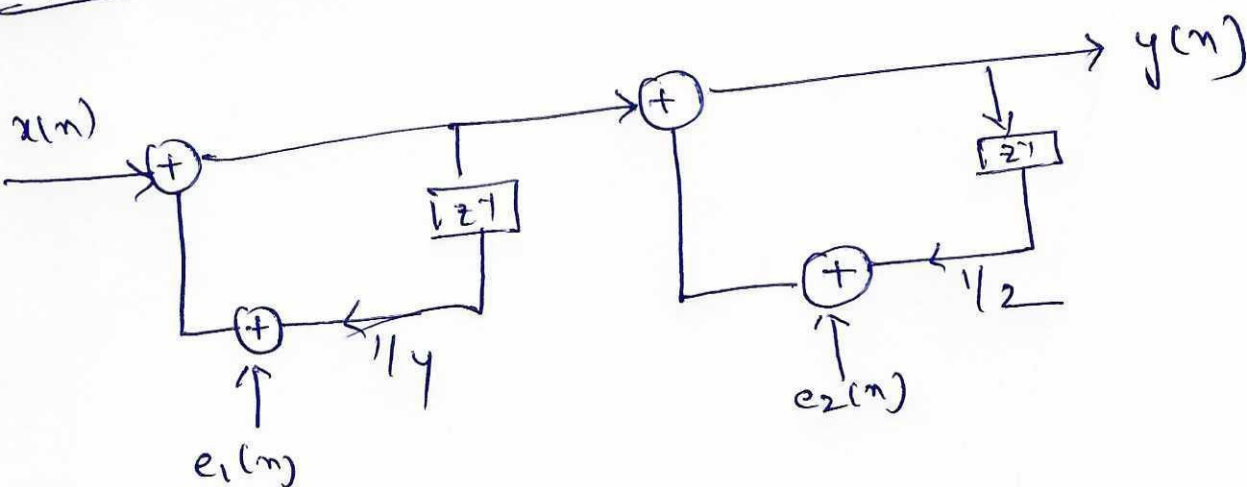
# I Cascade realization

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Try for parallel realization

# II Cascade realization



Variance o/p Noise

$$\sigma_{y_1}^2 = \sigma_e^2 \left[ \sum_{n=0}^{\infty} h_1^2(n) + \sum_{n=0}^{\infty} h_2^2(n) \right] \approx 2.90 \sigma_e^2$$

1st case

$$\sigma_{y_2}^2 = \sigma_e^2 \left[ \sum_{n=0}^{\infty} h_1^2(n) + \sum_{n=0}^{\infty} h_2^2(n) \right] \approx 3.16 \sigma_e^2$$

2nd case

$$\sum h_1^2(n) = 4/3$$

$$\sum h_2^2(n) = \frac{4}{1-1/4} - \frac{4}{1-1/8} + \frac{1}{1-1/16} = 1.83$$

$$\sum h_2^2(n) = 16/15$$

$$\frac{\sigma_{y_2}^2}{\sigma_{y_1}^2} = 1.09$$

Noise power in 2nd realization is 9% larger than in first realization.